Life Expectancy, Obesity and Physical Inactivity: A Relational Database

Course project, Databases (Erasmus+) · Nov 12, 2026 · @Bunda Rares

1. Introduction and description of the problem domain

This project builds a PostgreSQL database that combines three open indicators for 192 countries over 2000–2022: life expectancy at birth, adult obesity and insufficient physical activity.

The problem domain is public health at country level. Life expectancy summarises the mortality of a population. Obesity (BMI ⩾ 30) and insufficient physical activity are two widely monitored risk factors for noncommunicable diseases. The questions the database is meant to answer are:

- Which countries have the longest life expectancy, and how did it change between 2000 and 2022?
- Where are obesity and physical inactivity highest, and where did they grow fastest?
- Do countries with higher obesity or inactivity have shorter or longer life expectancy, and does this differ by WHO region or income group?
- Do men and women differ in inactivity and obesity?

The three indicators come from two organizations (WHO and the World Bank) in different file formats and layouts. They can be joined only through a shared country identifier (the ISO3 code) and the year. Designing a database that does this joining cleanly, instead of storing the files as they are, is the core of the project.

The results describe associations between country-level averages. They do not show that one indicator causes another, and the WHO values are modelled estimates, not raw measurements.

Project at a glance	
Countries	192 (present in all three sources)
Years	2000–2022 (23 years)
Tables	10 (9 data tables and an audit log)

Project at a glance	
Observations	30,912
SQL queries	17
DBMS	PostgreSQL 18
Repository	github.com/raresbunda0-alt/life-health-db

2. Description of the open-data source

The database uses three datasets from two recognized organizations. All files were downloaded on 11 November 2026 and stored unchanged in `data/raw/` of the repository.

	A. Physical inactivity	B. Obesity	C. Life expectancy
Dataset	Physical activity, insufficient, among adults aged 18+ years, prevalence (age-standardized estimate) (%)	Obesity among adults, BMI ⩾ 30, prevalence (crude estimate) (%)	Life expectancy at birth, total (years)
Indicator code	NCD_PAA	NCD_BMI_30C	SP.DYN.LE00.IN
Organization	World Health Organization, Global Health Observatory (GHO)	World Health Organization, Global Health Observatory (GHO)	World Bank, World Development Indicators
URL	https://ghoapi.azureedge.net/api/NCD_PAA	https://ghoapi.azureedge.net/api/NCD_BMI_30C	https://data.worldbank.org/indicator/SP.DYN.LE00.IN
Description	Share of adults who do less than 150 minutes of moderate or 75 minutes of vigorous activity per week (or equivalent), by country, year, sex and age group	Share of adults aged 18+ with a body mass index of 30 or more, by country, year and sex	Years a newborn would live if current mortality patterns stayed the same, by country and year
Main attributes	IndicatorCode, SpatialDimType, SpatialDim (ISO3), ParentLocationCode, ParentLocation, TimeDim, Dim1 (sex), Dim2 (age group), Value, NumericValue, Low, High, Date	Same structure as A	Country Name, Country Code, Indicator Name, Indicator Code, one column per year (1960–2025); separate metadata file with Region and IncomeGroup
Format	JSON (OData API)	JSON (OData API)	CSV (3 files: data, country metadata, indicator metadata)

	A. Physical inactivity	B. Obesity	C. Life expectancy
Volume in the source	41,262 rows, 25.3 MB, years 2000–2022	28,350 rows, 17.5 MB, years 1980–2024	266 rows × 66 year columns, 242 KB (country metadata 65 KB)
Volume used	13,248 rows	13,248 rows	4,416 values
License	CC BY-NC-SA 3.0 IGO	CC BY-NC-SA 3.0 IGO	CC BY 4.0
Retrieved on	11 Nov 2026	11 Nov 2026	11 Nov 2026 (file last updated by the World Bank on 2026-10-08)

doThe WHO licence allows copying and adapting the data for non-commercial purposes if WHO is credited, and derived work must carry the same or a similar licence (licence text, WHO data terms). This coursework project is non-commercial. WHO asks for the citation form: *World Health Organization. [Indicator]. Global Health Observatory. [URL], accessed 11 November 2026.*

The two WHO series differ in how they are standardized: physical inactivity is age-standardized and obesity is a crude estimate. Their numbers are not directly comparable, and the report says so wherever a query uses both.

3. Analysis of the source dataset

The analysis was done with the script `src/analyze_sources.py`, which reads the three sources and prints the counts below. The source files cannot be stored directly as database tables: they mix countries with aggregates, use different layouts and contain several levels of detail the project does not need.

3.1 Structure and size

	WHO physical inactivity	WHO obesity (crude)	World Bank life expectancy
Rows	41,262	28,350	266 (one per country or aggregate)
Layout	one row per place, year, sex and age group	one row per place, year, sex and age group	one row per country, one column per year (wide)

	WHO physical inactivity	WHO obesity (crude)	World Bank life expectancy
Years	2000–2022 (23)	1980–2024 (45)	1960–2025 (66 columns)
Countries	196	199	217 (plus 48 aggregates)
Sex values	male 18,262; female 18,262; both 4,738	male, female and both sexes, 9,450 each	not split (total only)
Age groups	18+ (14,214), 60–69, 70–79, 80+ (9,016 each)	18+ only	not applicable
Value range	2.39 to 96.67 %	0.05 to 80.89 %	years
Missing values	0	0	251 empty cells of 14,322 country-year cells (1.75 %)
Duplicates	0	0	not applicable

3.2 Meaning of the important attributes

- **SpatialDim / Country Code**: the ISO3 code of the country. This is the identifier shared by all three sources.
- **SpatialDimType**: tells whether a WHO row describes a country, a WHO region, a World Bank income group or the whole world.
- **ParentLocationCode / ParentLocation**: the WHO region of the country (AFR, AMR, EMR, EUR, SEAR, WPR).
- **TimeDim**: the year. **Dim1**: the sex (SEX_BTSX both, SEX_MLE male, SEX_FMLE female). **Dim2**: the age group.
- **NumericValue, Low, High**: the estimate and the lower and upper bound of its interval. **Value** is the same information as text, for example `17.3 [5.5-36.4]`, so it is not used.
- **Region, IncomeGroup** (World Bank country metadata): the region and income class of each country. Aggregates have no region.

3.3 Problems found in the source data

1. **Non-country rows in the WHO files.** 759 rows in the inactivity file and 1,485 in the obesity file describe WHO regions, World Bank income groups or the world (inactivity: 414 region, 276 income-group and 69 global rows; obesity: 810, 540 and 135). They have no WHO region code.
2. **Several age groups.** The inactivity file contains four age groups; only 18+ matches the chosen indicator.
3. **No country names in the WHO files.** Only ISO3 codes are given. Names come from the World Bank metadata.
4. **Wide format in the World Bank file.** Years are columns, and the file starts with four non-data lines before the header and ends every line with an extra comma.
5. **Aggregates in the World Bank file.** 48 entries such as World or income groups have no region in the metadata and must be removed.
6. **Countries missing in one source.** WHO has COK and NIU (both WHO files) and TKL (obesity only), which the World Bank file does not contain. The World Bank file has 21 territories that the WHO obesity file lacks, for example ABW, HKG, MAC and XKX.
7. **Different time coverage.** The three sources cover 2000–2022, 1980–2024 and 1960–2025.
8. **Different kinds of estimate.** Inactivity is age-standardized, obesity is crude, and both are modelled by WHO. Life expectancy has no sex split.

3.4 Units, categories, time, space and relationships

- **Units:** WHO indicators are percentages of the adult population; life expectancy is in years.
- **Categorical attributes:** sex, age group, location type, WHO region, World Bank region, income group.
- **Temporal attribute:** the year (a calendar year, no month or day).
- **Spatial attribute:** the country, with two different regional groupings (WHO regions and World Bank regions). There are no coordinates.
- **Relationships:** a country belongs to one WHO region, one World Bank region and one income group; a country has many observations, one per indicator, sex and year.

3.5 Selected subset

The database keeps the **192 countries present in all three sources**, the **years 2000–2022** (the period where all three overlap), **adults 18+**, and for the WHO indicators all three sex values. This gives 192 × 23 × 3 = 13,248 rows for each WHO indicator and 192 × 23 = 4,416 for life expectancy, 30,912 observations in total. The subset was chosen so that every country-year can be compared across all three indicators, which makes the analysis queries consistent.

4. Conceptual database model

The model has ten entities. The central entity is the observation: one measured value of one indicator, for one country, one sex and one year. Countries are described once and linked to their regions and income group, so this information is not repeated on every measurement.

ER diagram. *(Insert here the screenshot* `docs/er_diagram.png`*, generated with pgAdmin, ERD For Database.)*

Entities

Entity	Meaning	Origin
Country	A country identified by its ISO3 code	Codes from all sources, names from World Bank metadata
WHO region	One of six WHO regions (AFR, AMR, EMR, EUR, SEAR, WPR)	WHO files (ParentLocation)
World Bank region	One of seven World Bank regions	World Bank country metadata
Income group	Low, lower middle, upper middle or high income	World Bank country metadata
Sex	Both sexes, male, female	Codes from WHO files; table designed by the students
Indicator	One of the three measured indicators with its unit	Codes and names from the sources; unit and structure by the students

Entity	Meaning	Origin
Observation	One value (with optional interval) per country, indicator, sex and year	Values and years from the sources; keys and links by the students
Data source	Description of a dataset: organization, URL, licence, retrieval date	Written by the students from the source pages
Import batch	One import run per dataset with row counts	Created by the import script
Audit log	Record of changed or deleted observation values	Created by the students (filled by a trigger)

Relationships

Relationship	Type	Participation
WHO region → Country	one-to-many	optional for the country
World Bank region → Country	one-to-many	optional for the country
Income group → Country	one-to-many	optional for the country
Country → Observation	one-to-many	mandatory
Indicator → Observation	one-to-many	mandatory
Sex → Observation	one-to-many	mandatory
Data source → Indicator	one-to-many	mandatory
Data source → Import batch	one-to-many	mandatory
Import batch → Observation	one-to-many	optional
Country ↔ Indicator	many-to-many, resolved by Observation (with sex, year and value as attributes)	mandatory

5. Logical relational model

The conceptual model is translated into ten relations. Every table has a surrogate integer primary key generated by the database. Natural identifiers such as the ISO3 code are kept as UNIQUE columns. The same schema is in `docs/logical_model.md` and `sql/01_schema.sql`.

Table	Primary key	Foreign keys	Other attributes
data_source	source_id	none	name, organization, url, license, retrieved_on

Table	Primary key	Foreign keys	Other attributes
import_batch	batch_id	source_id → data_source	started_at, rows_read, rows_loaded, rows_rejected
who_region	who_region_id	none	code, name
wb_region	wb_region_id	none	name
income_group	income_group_id	none	name
country	country_id	who_region_id → who_region; wb_region_id → wb_region; income_group_id → income_group	iso3, name
sex	sex_id	none	code, label
indicator	indicator_id	source_id → data_source	code, name, unit
observation	observation_id	country_id → country; indicator_id → indicator; sex_id → sex; batch_id → import_batch	year, value, ci_low, ci_high
audit_log	audit_id	none	table_name, operation, record_id, old_value, new_value, changed_at, changed_by

Candidate keys. `country.iso3`, `who_region.code`, `sex.code`, `indicator.code` and the names of the other lookup tables are UNIQUE. In `observation` the combination (country_id, indicator_id, sex_id, year) is UNIQUE, so a country cannot have two values for the same indicator, sex and year.

Optional and mandatory. The three region and income foreign keys of `country`, the interval columns `ci_low` and `ci_high`, and `batch_id` are optional. Most other columns are NOT NULL. The exceptions are the row counts of an import batch, the old and new values of the audit log, and `data_source.license`.

6. Normalization and design decisions

The schema is in third normal form (3NF).

- **1NF:** every column holds one atomic value. The WHO text value `17.3 [5.5-36.4]` is split into `value`, `ci_low` and `ci_high`, and the World Bank year columns become one row per year.
- **2NF:** every attribute depends on the whole key. All tables have a single-column surrogate key, and the natural key of `observation` (country, indicator, sex, year) determines the value and its interval as a whole. The unit of measurement depends only on the indicator, so it is stored in `indicator`, not in `observation`.
- **3NF:** no attribute depends on another non-key attribute. Country name, regions and income group depend on the country and are stored once in `country` and the lookup tables, not on 30,912 observation rows.

Design decisions

1. **One generic `observation` table instead of one table per indicator.** A new indicator needs only a new row in `indicator`, and all indicators share the same keys, constraints and indexes. The cost is that the meaning of `value` depends on the indicator, so range rules (a percentage cannot exceed 100) need a trigger, and the indicators are placed side by side by the view `v_country_year`.
2. **WHO regions and World Bank regions are separate tables.** The two organizations group countries differently, for example WHO counts Israel in its European Region, which is an administrative and not a geographic grouping. Merging them would lose one of the groupings.
3. **Sex is a lookup table.** Life expectancy is stored for both sexes only; a trigger enforces this.
4. **Surrogate keys plus UNIQUE natural keys.** Integer keys keep joins small, while `iso3`, `code` and the observation key stay unique.
5. **Source tracking inside the database.** `data_source`, `indicator.source_id` and `import_batch` let every value be traced to its dataset and import run.
6. **Interval columns are optional.** WHO provides a lower and upper bound; the World Bank does not, so `ci_low` and `ci_high` are NULL for life expectancy.
7. **Age group is not stored.** Only the 18+ group is kept. If other age groups were needed, an `age_group` table would be added.

Deliberate redundancy

The only deliberate redundancy is `import_batch.rows_loaded`, which repeats the number of observation rows of a batch. It is kept as a record of what the import loaded at that time, and the procedure `refresh_batch_stats` recalculates it from the data. The view `v_country_year` is a presentation of existing data and stores nothing.

7. Description of tables and attributes

The column *Origin* separates data that come from the external sources (**source**) from structures and values introduced by the students (**added**). Primary keys are integers generated by the database (`GENERATED ALWAYS AS IDENTITY`).

Table	Column	Type	Constraints	Origin
data_source	source_id	INT	PRIMARY KEY	added
data_source	name	VARCHAR(200)	NOT NULL, UNIQUE	added
data_source	organization	VARCHAR(200)	NOT NULL	added
data_source	url	TEXT	NOT NULL	added
data_source	license	VARCHAR(300)	optional	added
data_source	retrieved_on	DATE	NOT NULL	added
import_batch	batch_id	INT	PRIMARY KEY	added
import_batch	source_id	INT	NOT NULL, FK → data_source, ON DELETE RESTRICT	added
import_batch	started_at	TIMESTAMP	NOT NULL, default now()	added
import_batch	rows_read, rows_loaded, rows_rejected	INT	CHECK ≥ 0	added
who_region	who_region_id	INT	PRIMARY KEY	added
who_region	code	VARCHAR(10)	NOT NULL, UNIQUE (for example EUR)	source

Table	Column	Type	Constraints	Origin
who_region	name	VARCHAR(100)	NOT NULL	source
wb_region	wb_region_id	INT	PRIMARY KEY	added
wb_region	name	VARCHAR(100)	NOT NULL, UNIQUE	source
income_group	income_group_id	INT	PRIMARY KEY	added
income_group	name	VARCHAR(100)	NOT NULL, UNIQUE	source
country	country_id	INT	PRIMARY KEY	added
country	iso3	CHAR(3)	NOT NULL, UNIQUE, CHECK three capital letters	source
country	name	VARCHAR(150)	NOT NULL	source (World Bank)
country	who_region_id	INT	FK → who_region, ON DELETE SET NULL, optional	source
country	wb_region_id	INT	FK → wb_region, ON DELETE SET NULL, optional	source
country	income_group_id	INT	FK → income_group, ON DELETE SET NULL, optional	source
sex	sex_id	INT	PRIMARY KEY	added
sex	code	VARCHAR(20)	NOT NULL, UNIQUE (SEX_BTSX, SEX_MLE, SEX_FMLE)	source
sex	label	VARCHAR(30)	NOT NULL (Both sexes, Male, Female)	added
indicator	indicator_id	INT	PRIMARY KEY	added

Table	Column	Type	Constraints	Origin
indicator	code	VARCHAR(50)	NOT NULL, UNIQUE	source
indicator	name	VARCHAR(300)	NOT NULL (shortened name)	added
indicator	unit	VARCHAR(20)	NOT NULL (% or years)	added
indicator	source_id	INT	NOT NULL, FK → data_source, ON DELETE RESTRICT	added
observation	observation_id	BIGINT	PRIMARY KEY	added
observation	country_id	INT	NOT NULL, FK → country, ON DELETE CASCADE	added
observation	indicator_id	INT	NOT NULL, FK → indicator, ON DELETE RESTRICT	added
observation	sex_id	INT	NOT NULL, FK → sex, ON DELETE RESTRICT	added
observation	year	SMALLINT	NOT NULL, CHECK between 1960 and 2100	source
observation	value	NUMERIC(8,3)	NOT NULL, CHECK ≥ 0	source
observation	ci_low, ci_high	NUMERIC(8,3)	optional, CHECK ≥ 0, and ci_low ≤ ci_high	source
observation	batch_id	INT	FK → import_batch, ON DELETE SET NULL, optional	added
observation	(country_id, indicator_id, sex_id, year)		UNIQUE	added

Table	Column	Type	Constraints	Origin
audit_log	audit_id	BIGINT	PRIMARY KEY	added
audit_log	table_name	VARCHAR(50)	NOT NULL	added
audit_log	operation	VARCHAR(10)	NOT NULL, CHECK UPDATE or DELETE	added
audit_log	record_id	BIGINT	NOT NULL	added
audit_log	old_value, new_value	NUMERIC(8,3)	optional (new_value is empty for a delete)	added
audit_log	changed_at	TIMESTAMP	NOT NULL, default now()	added
audit_log	changed_by	TEXT	NOT NULL, default current_user	added

Rows after the import

Table	Rows
data_source	3
import_batch	3
who_region	6
wb_region	7
income_group	4
country	192
sex	3
indicator	3
observation	30,912

The `audit_log` table is empty after the import. It is filled only when an observation value is changed or deleted.

8. Data import and transformation procedure

The import is done by the Python script `src/import_data.py` (library psycopg). It follows the pipeline *source files → validation → cleaning → transformation → relational mapping → database* and can be run again on an empty or filled database to recreate the same content.

Step	What the script does
1. Read	Reads the two WHO JSON files and the World Bank CSV files. For the World Bank file it skips the four non-data lines before the header.
2. Validate and clean the WHO data	Keeps only rows of type COUNTRY, age group 18+, years 2000–2022 and with a numeric value. Skips duplicates of (country, sex, year), negative values and rows where the lower bound is above the upper bound.
3. Clean the World Bank data	Uses the country metadata to remove the 48 aggregates (World, regions, income groups). Empty year cells are skipped.
4. Select the subset	Keeps the countries present in all three sources: 192.
5. Transform	Converts the wide World Bank table (one column per year) to one row per country and year; rounds values to three decimals; takes country names from the World Bank metadata and the WHO region from the WHO field ParentLocationCode.
6. Map identifiers	Loads the lookup tables first (sources, indicators, regions, income groups, countries) and then replaces ISO3 codes, indicator codes and sex codes with the generated integer keys of those tables.
7. Load	Empties the data tables (`TRUNCATE ... RESTART IDENTITY CASCADE`), inserts all rows in one transaction and records one `import_batch` row per dataset.

Step	What the script does
8. Report	Prints rows read and loaded per dataset and the row count of every table.

The database password is typed when the script starts and is not stored in the code. The retrieval date (11 November 2026) is stored in `data_source`.

Import result

Dataset	Rows read	Rows loaded	Rows not loaded	Share loaded
WHO physical inactivity	41,262	13,248	28,014	32.1 %
WHO obesity	28,350	13,248	15,102	46.7 %
World Bank life expectancy	14,322	4,416	9,906	30.8 %

The rows that were not loaded were removed on purpose: regions and other aggregates, other age groups, years outside 2000–2022 and countries that are not in all three sources. Both WHO files loaded exactly 192 countries × 23 years × 3 sexes = 13,248 rows and life expectancy exactly 192 × 23 = 4,416 values. No expected country-year was lost, so no row inside the subset was rejected.

Order to recreate the database

1. `sql/01_schema.sql` creates the tables and the three sex values.
2. `python src/import_data.py` loads the data.
3. `sql/02_views_indexes.sql` creates the view and the indexes.
4. `sql/04_functions_triggers.sql` creates the functions, the procedure, the triggers and the audit table.

The complete instructions are in the README of the repository.

9. Implemented constraints

The database rejects invalid data at database level. The constraints are defined in `sql/01_schema.sql`, and the rules that need data from other tables are implemented by a trigger (section 10).

Kind of integrity	How it is implemented
Entity integrity	Every table has a PRIMARY KEY (generated integer).
Referential integrity	FOREIGN KEY constraints between all related tables.
Domain constraints	Data types (CHAR(3), SMALLINT, NUMERIC(8,3), DATE), NOT NULL, and CHECK constraints such as three capital letters for `iso3` and `operation` limited to UPDATE or DELETE.
Valid ranges	`year` between 1960 and 2100; `value`, `ci_low`, `ci_high` not negative; `ci_low` not above `ci_high`; trigger: a percentage not above 100 and life expectancy not above 120 years.
Uniqueness	UNIQUE on ISO3 codes, codes of lookup tables, source names, and on (country, indicator, sex, year) in `observation`.
Missing values	Mandatory columns are NOT NULL; region, income group, interval and batch columns are optional because the sources do not always provide them.

Deletion behaviour

Deleting a row in	Effect on rows that refer to it	Reason
country	its observations are deleted (CASCADE)	an observation has no meaning without its country
indicator, sex	deletion is refused while observations exist (RESTRICT)	protects the measurements
data_source	deletion is refused while indicators or batches exist (RESTRICT)	keeps the origin of the data
import_batch	`batch_id` of its observations becomes NULL (SET NULL)	the values stay, only the log entry is lost

Deleting a row in	Effect on rows that refer to it	Reason
who_region, wb_region, income_group	`country` references become NULL (SET NULL)	a country stays valid without its grouping

Primary keys are generated and never changed, so updates of keys are not needed.

Integrity tests

The script `sql/05_integrity_tests.sql` contains eleven statements. Nine try to store invalid data and must fail; two are demonstrations that undo themselves. After running all of them, the data check still returned 30,912 observations.

Test	Statement	Rule that stops it	Result
T1	insert a copy of an existing observation	UNIQUE (country, indicator, sex, year)	rejected: duplicate key
T2	insert value −5	CHECK value ⩾ 0	rejected: check constraint violated
T3	insert with year NULL	NOT NULL	rejected: null value not allowed
T4	insert with country_id 999999	FOREIGN KEY to country	rejected: foreign key violated
T5	insert ci_low 30 and ci_high 20	CHECK ci_low ⩽ ci_high	rejected: check constraint violated
T6	insert country code `ro1`	CHECK three capital letters	rejected: check constraint violated
T7	delete the indicator NCD_PAA	ON DELETE RESTRICT	rejected: observations still refer to it
T8	insert a percentage of 150	trigger	rejected: *Percentage indicator NCD_PAA cannot exceed 100 (got 150)*
T9	insert male life expectancy	trigger	rejected: *Life expectancy is stored for both sexes only (got SEX_MLE)*

Test	Statement	Rule that stops it	Result
T10	delete Romania inside a block that is rolled back	ON DELETE CASCADE and audit trigger	shows the cascade and audit rows, then undoes everything
T11	change one value inside a block that is rolled back	audit trigger	shows the audit row, then undoes everything

(Insert here screenshots of the error messages of T1 to T9 and the messages of T10 and T11 from pgAdmin.)

10. Views, indexes, procedures/functions and triggers

These objects are created by `sql/02_views_indexes.sql` and `sql/04_functions_triggers.sql`. They are introduced by the students and contain no external data.

Object	Type	Purpose
v_country_year	view	One row per country and year with life expectancy, obesity and inactivity side by side (for both sexes, male and female), plus the WHO region, World Bank region and income group.
idx_obs_indicator_year	index on observation (indicator_id, year)	Speeds up filters by indicator and year, which almost every query uses.
idx_country_who_region, idx_country_wb_region, idx_country_income	indexes on country	Speed up grouping and joining by region and income group.
get_country_trend(iso3)	function	Returns the yearly life expectancy, obesity and inactivity of one country.

Object	Type	Purpose
pct_change(old, new)	function	Percentage change between two values; NULL if the old value is missing or zero.
refresh_batch_stats()	procedure	Recalculates `rows_loaded` and `rows_rejected` of every import batch from the stored observations.
trg_observation_range	trigger (before insert or update on observation)	Enforces rules that need other tables: percentages not above 100, life expectancy not above 120 years, life expectancy only for both sexes.
trg_observation_audit	trigger (after update or delete on observation)	Writes the old and new value, the user and the time of every changed or deleted value to `audit_log`.

Details

The view. The three indicators are stored as rows of one table, so a country-year with all three values needs a pivot. The view does this once, with conditional aggregation (`MAX(CASE WHEN indicator = ... THEN value END)`), and the queries in section 11 read from it.

The indexes. PostgreSQL also creates indexes automatically for every primary key and UNIQUE constraint. The unique index on (country_id, indicator_id, sex_id, year) already supports lookups of one country.

The functions. `SELECT * FROM get_country_trend('ROU');` returns 23 rows for Romania. `pct_change(12.783, 35.644)` returns 178.8, the relative growth of Romanian obesity from 2000 to 2022.

The triggers. The range trigger raises the errors shown in tests T8 and T9. The audit trigger fills the table `audit_log`, which is the tenth table of the database and is shown working in test T11.

11. SQL queries and interpretation of results

The 17 queries are in `sql/03_queries.sql`. They read from the view `v_country_year`, which already joins `observation`, `country`, `indicator`, `sex`, the two region tables and `income_group`. Averages are simple averages over countries, not weighted by population. The inactivity values are age-standardized and the obesity values are crude estimates, so they are compared only in direction, not in size. Long results are cut to the first rows, and the cut is stated.

Required element	Queries
Filtering and sorting	Q1, Q2, Q8
JOIN of several tables	Q5, Q6, Q15, Q16 (and the view, which joins six tables)
Aggregate functions and GROUP BY	Q3, Q4, Q7, Q10, Q12, Q13
HAVING	Q4, Q12
Subqueries and CTEs	Q9 (subquery), Q10, Q11, Q13, Q14, Q16 (CTEs)
Calculations on stored data	Q5, Q6, Q8, Q10, Q14
Date and time operations	Q14 (change from the previous year), Q15 (days since download)
Complex analytical query	Q16

Q1. Highest life expectancy

Question. Which 10 countries had the highest life expectancy in 2022?

```
SELECT country, who_region, life_expectancy
FROM v_country_year
WHERE year = 2022
ORDER BY life_expectancy DESC
LIMIT 10;
```

country	who_region	life_expectancy
Andorra	Europe	84.016
Japan	Western Pacific	83.996
Switzerland	Europe	83.605
Australia	Western Pacific	83.200
Spain	Europe	83.134
Sweden	Europe	83.059
Luxembourg	Europe	82.946
Singapore	Western Pacific	82.895
Israel	Europe	82.700
Italy	Europe	82.700

Interpretation. Andorra (84.0 years) and Japan (84.0) lead the ranking. Seven of the ten countries are in the WHO European Region and three are in the Western Pacific Region. Israel appears under Europe because WHO assigns it to its European Region, which is an administrative grouping and not a geographic one.

Q2. High obesity together with high inactivity

Question. Which countries in 2022 had obesity of at least 25 % and physical inactivity of at least 35 %?

```
SELECT country, who_region, obesity, inactivity
FROM v_country_year
WHERE year = 2022 AND obesity >= 25 AND inactivity >= 35
ORDER BY obesity DESC;
```

The query returns 40 rows; the first 10 are shown.

country	who_region	obesity	inactivity
Naoero	Western Pacific	70.798	35.451
Bahamas, The	Americas	47.097	37.167

country	who_region	obesity	inactivity
St. Kitts and Nevis	Americas	46.497	41.695
Belize	Americas	41.708	41.519
Qatar	Eastern Mediterranean	41.273	53.483
Kuwait	Eastern Mediterranean	40.992	63.315
Puerto Rico (US)	Americas	40.825	41.552
Chile	Americas	39.202	38.214
Saudi Arabia	Eastern Mediterranean	38.300	51.536
Barbados	Americas	38.255	44.033

Interpretation. 40 countries meet both thresholds: 23 in the Americas, 10 in the Eastern Mediterranean, 5 in Europe, and one each in the Western Pacific and Africa. Nauru has by far the highest obesity (70.8 %). The United Arab Emirates (66.1 %) and Kuwait (63.3 %) have the highest inactivity among them. Romania is on the list as well (35.6 % obesity, 36.8 % inactivity).

Q3. Life expectancy by region and income group

Question. What was the average life expectancy in 2022 for each combination of WHO region and income group?

```
SELECT who_region, income_group, COUNT(*) AS countries,
       ROUND(AVG(life_expectancy), 1) AS avg_life_expectancy
FROM v_country_year
WHERE year = 2022
GROUP BY who_region, income_group
ORDER BY who_region, avg_life_expectancy DESC;
```

who_region	income_group	countries	avg_life_expectancy
Africa	High income	1	74.5
Africa	Upper middle income	7	70.1
Africa	Lower middle income	20	63.5

who_region	income_group	countries	avg_life_expectancy
Africa	Low income	19	60.5
Americas	High income	13	76.4
Americas	Upper middle income	18	73.8
Americas	Lower middle income	5	70.2
Eastern Mediterranean	High income	6	79.9
Eastern Mediterranean	Upper middle income	4	75.1
Eastern Mediterranean	Lower middle income	7	72.8
Eastern Mediterranean	Low income	5	65.2
Europe	High income	34	80.1
Europe	Upper middle income	14	74.6
Europe	Lower middle income	3	71.9
South-East Asia	Upper middle income	3	77.8
South-East Asia	Low income	1	73.6
South-East Asia	Lower middle income	6	70.4
Western Pacific	High income	8	77.3
Western Pacific	Upper middle income	12	71.0
Western Pacific	Lower middle income	6	68.7

Interpretation. Within almost every region, life expectancy falls as the income group gets lower. In Africa it goes from 74.5 years (high income) to 60.5 (low income). The one exception is South-East Asia, where the single low-income country (73.6) is above the lower-middle group (70.4). The highest average is in high-income European countries (80.1) and the lowest in low-income African countries (60.5), a gap of 19.6 years. Two groups contain one country each (high-income Africa and low-income South-East Asia), so their averages describe a single country.

Q4. Regions with high inactivity

Question. Which WHO regions had an average physical inactivity above 30 % in 2022?

```
SELECT who_region, COUNT(*) AS countries, ROUND(AVG(inactivity), 1) AS
avg_inactivity
FROM v_country_year
WHERE year = 2022
GROUP BY who_region
HAVING AVG(inactivity) > 30
ORDER BY avg_inactivity DESC;
```

who_region	countries	avg_inactivity
Eastern Mediterranean	22	39.4
Americas	36	37.8

Interpretation. Only two regions exceed 30 %: the Eastern Mediterranean (39.4 % on average, 22 countries) and the Americas (37.8 %, 36 countries). The HAVING clause removes the other four regions.

Q5. Largest gains in life expectancy

Question. Which 10 countries gained the most years of life expectancy between 2000 and 2022?

```
SELECT b.country, b.who_region,
       a.life_expectancy AS le_2000, b.life_expectancy AS le_2022,
       ROUND(b.life_expectancy - a.life_expectancy, 1) AS gain_years
FROM v_country_year a
JOIN v_country_year b ON b.country_id = a.country_id
WHERE a.year = 2000 AND b.year = 2022
ORDER BY gain_years DESC
LIMIT 10;
```

country	who_region	le_2000	le_2022	gain_years
Malawi	Africa	46.170	66.037	19.9
Rwanda	Africa	47.798	67.528	19.7
Zambia	Africa	46.577	65.279	18.7
Uganda	Africa	49.640	67.675	18.0

country	who_region	le_2000	le_2022	gain_years
Angola	Africa	46.501	64.246	17.7
Botswana	Africa	52.263	68.749	16.5
Sierra Leone	Africa	44.834	61.275	16.4
Zimbabwe	Africa	46.034	62.360	16.3
Ethiopia	Africa	50.880	66.897	16.0
Eswatini	Africa	47.627	63.028	15.4

Interpretation. All ten countries are in the WHO African Region, with gains of 15.4 to 19.9 years (Malawi is first). They started low in 2000 (44.8 to 52.3 years) and reached 61.3 to 68.7 years in 2022, so this is largely a catch-up from a very low starting point. The database shows what changed but not why.

Q6. Largest increases in obesity

Question. In which 10 countries did adult obesity increase the most between 2000 and 2022?

```
SELECT b.country, b.who_region,
       a.obesity AS obesity_2000, b.obesity AS obesity_2022,
       ROUND(b.obesity - a.obesity, 1) AS increase_points
FROM v_country_year a
JOIN v_country_year b ON b.country_id = a.country_id
WHERE a.year = 2000 AND b.year = 2022
ORDER BY increase_points DESC
LIMIT 10;
```

country	who_region	obesity_2000	obesity_2022	increase_points
Romania	Europe	12.783	35.644	22.9
Brunei Darussalam	Western Pacific	13.634	35.487	21.9
Panama	Americas	16.544	35.938	19.4
Tonga	Western Pacific	51.846	70.800	19.0
Chile	Americas	20.362	39.202	18.8

country	who_region	obesity_2000	obesity_2022	increase_points
St. Kitts and Nevis	Americas	27.889	46.497	18.6
Bahamas, The	Americas	28.526	47.097	18.6
Georgia	Europe	20.088	38.443	18.4
Uzbekistan	Europe	10.119	28.285	18.2
Samoa	Western Pacific	43.985	62.119	18.1

Interpretation. Romania has the largest increase (22.9 percentage points, from 12.8 % to 35.6 %), followed by Brunei (21.9) and Panama (19.4). The list contains four countries of the Americas, three of Europe and three of the Western Pacific. Tonga and Samoa have the highest obesity levels in the group (70.8 % and 62.1 %).

Q7. Correlation between obesity and life expectancy

Question. How strongly are obesity and life expectancy correlated in each WHO region in 2022?

```
SELECT who_region, COUNT(*) AS countries,
       ROUND(CORR(obesity, life_expectancy)::numeric, 2) AS correlation
FROM v_country_year
WHERE year = 2022
GROUP BY who_region
ORDER BY correlation;
```

who_region	countries	correlation
Western Pacific	26	−0.46
Europe	51	−0.33
Africa	47	0.31
Americas	36	0.32
South-East Asia	10	0.75
Eastern Mediterranean	22	0.75

Interpretation. The correlation is negative in the Western Pacific (−0.46) and Europe (−0.33), weak and positive in Africa and the Americas (about 0.3), and strong and positive in South-East Asia and the Eastern Mediterranean (0.75). The sign changes between regions, so there is no single global relationship. South-East Asia has only 10 countries, so its value is less reliable. A positive correlation does not mean that obesity lengthens life: where incomes differ widely, richer countries tend to have both higher obesity and longer life expectancy, an explanation this query does not test.

Q8. Gender gap in inactivity

Question. In which 10 countries are women much more physically inactive than men in 2022?

```
SELECT country, who_region, inactivity_male, inactivity_female,
       ROUND(inactivity_female - inactivity_male, 1) AS gap_points
FROM v_country_year
WHERE year = 2022
ORDER BY gap_points DESC
LIMIT 10;
```

country	who_region	inactivity_male	inactivity_female	gap_points
Afghanistan	Eastern Mediterranean	20.019	46.025	26.0
Pakistan	Eastern Mediterranean	34.140	57.008	22.9
Cuba	Americas	50.245	72.030	21.8
Guyana	Americas	24.214	45.069	20.9
Iran, Islamic Rep.	Eastern Mediterranean	35.915	56.650	20.7
Bahamas, The	Americas	26.575	46.624	20.0
Kiribati	Western Pacific	21.242	41.143	19.9
Iraq	Eastern Mediterranean	41.894	61.510	19.6
Trinidad and Tobago	Americas	34.634	54.167	19.5

country	who_region	inactivity_male	inactivity_female	gap_points
Palau	Western Pacific	16.579	35.967	19.4

Interpretation. Women are more inactive than men in all ten countries, by 19.4 to 26.0 percentage points. Afghanistan has the largest gap (20.0 % of men against 46.0 % of women), followed by Pakistan (22.9) and Cuba (21.8). Four countries are in the Eastern Mediterranean, four in the Americas and two in the Western Pacific.

Q9. Countries above their region's average obesity

Question. Which countries have obesity above the average of their own WHO region in 2022, and by how much?

```
SELECT v.country, v.who_region, v.obesity,
       (SELECT ROUND(AVG(x.obesity), 1) FROM v_country_year x
        WHERE x.who_region = v.who_region AND x.year = 2022) AS region_avg
FROM v_country_year v
WHERE v.year = 2022
  AND v.obesity > (SELECT AVG(x.obesity) FROM v_country_year x
                   WHERE x.who_region = v.who_region AND x.year = 2022)
ORDER BY v.obesity DESC
LIMIT 15;
```

The first 15 rows are shown.

country	who_region	obesity	region_avg
Tonga	Western Pacific	70.800	29.3
Naoero	Western Pacific	70.798	29.3
Tuvalu	Western Pacific	63.558	29.3
Samoa	Western Pacific	62.119	29.3
Bahamas, The	Americas	47.097	32.0
St. Kitts and Nevis	Americas	46.497	32.0
Micronesia, Fed. Sts.	Western Pacific	45.744	29.3
Kiribati	Western Pacific	44.675	29.3

country	who_region	obesity	region_avg
Palau	Western Pacific	41.916	29.3
Egypt, Arab Rep.	Eastern Mediterranean	41.875	27.8
Belize	Americas	41.708	32.0
United States	Americas	41.298	32.0
Qatar	Eastern Mediterranean	41.273	27.8
Marshall Islands	Western Pacific	41.266	29.3
Kuwait	Eastern Mediterranean	40.992	27.8

Interpretation. The countries furthest above their region's average are mostly Pacific island states: Tonga has 70.8 % obesity against a regional average of 29.3 %. In the Americas, the Bahamas, St Kitts and Nevis, Belize and the United States are above the average of 32.0 %. In the Eastern Mediterranean, Egypt, Qatar and Kuwait are above 27.8 %. The regional averages count every country equally, so they do not reflect population size.

Q10. Obesity and life expectancy trends by income group

Question. For each income group, in how many countries did obesity rise, life expectancy rise, or both rise between 2000 and 2022?

```
WITH a AS (SELECT * FROM v_country_year WHERE year = 2000),
     b AS (SELECT * FROM v_country_year WHERE year = 2022)
SELECT b.income_group, COUNT(*) AS countries,
       SUM(CASE WHEN b.obesity > a.obesity THEN 1 ELSE 0 END) AS
obesity_rose,
       SUM(CASE WHEN b.life_expectancy > a.life_expectancy THEN 1 ELSE 0 END)
AS life_exp_rose,
       SUM(CASE WHEN b.obesity > a.obesity AND b.life_expectancy >
a.life_expectancy
                THEN 1 ELSE 0 END) AS both_rose
FROM a JOIN b ON b.country_id = a.country_id
GROUP BY b.income_group
ORDER BY b.income_group;
```

income_group	countries	obesity_rose	life_exp_rose	both_rose
High income	62	61	61	60
Low income	25	25	24	24
Lower middle income	47	47	47	47
Upper middle income	58	58	57	57

Interpretation. Obesity rose in 191 of the 192 countries, life expectancy rose in 189, and both rose in 188. The share is nearly the same in every income group, so income does not separate the countries here. Because both indicators rose almost everywhere in the same period, the result shows that the two trends happened together, not that one caused the other.

Q11. Most inactive countries per income group

Question. Which are the three most physically inactive countries in each income group in 2022?

```
WITH ranked AS (
    SELECT income_group, country, inactivity,
           RANK() OVER (PARTITION BY income_group ORDER BY inactivity DESC)
AS rnk
    FROM v_country_year
    WHERE year = 2022
)
SELECT * FROM ranked WHERE rnk <= 3 ORDER BY income_group, rnk;
```

income_group	country	inactivity	rnk
High income	United Arab Emirates	66.138	1
High income	Kuwait	63.315	2
High income	Korea, Rep.	58.064	3
Low income	Syrian Arab Republic	37.840	1
Low income	Afghanistan	33.356	2
Low income	Mali	29.820	3

income_group	country	inactivity	rnk
Lower middle income	Lebanon	58.616	1
Lower middle income	India	49.418	2
Lower middle income	Venezuela, RB	45.729	3
Upper middle income	Cuba	61.076	1
Upper middle income	Iraq	52.022	2
Upper middle income	Suriname	48.855	3

Interpretation. The highest inactivity is found among high-income countries: the United Arab Emirates (66.1 %), Kuwait (63.3 %) and South Korea (58.1 %). The top three in the low-income group are much lower (Syria 37.8 %, Afghanistan 33.4 %, Mali 29.8 %). Lebanon leads the lower-middle group (58.6 %) and Cuba the upper-middle group (61.1 %).

Q12. Completeness of the data per region

Question. How many countries per WHO region have values for all three indicators in all 23 years?

```
SELECT who_region, COUNT(*) AS complete_countries
FROM (
    SELECT country_id, who_region
    FROM v_country_year
    WHERE life_expectancy IS NOT NULL AND obesity IS NOT NULL AND inactivity
IS NOT NULL
    GROUP BY country_id, who_region
    HAVING COUNT(*) = 23
) t
GROUP BY who_region
ORDER BY who_region;
```

who_region	complete_countries
Africa	47
Americas	36

who_region	complete_countries
Eastern Mediterranean	22
Europe	51
South-East Asia	10
Western Pacific	26

Interpretation. All 192 countries are complete (47 + 36 + 22 + 51 + 10 + 26). This is expected, because the subset was built from the countries present in all three sources. The query works as a check of the import and not as an analytical finding.

Q13. Life expectancy by level of inactivity

Question. Is life expectancy different in countries with low, medium and high physical inactivity in 2022?

```
WITH y AS (
    SELECT life_expectancy,
           CASE WHEN inactivity < 20 THEN '1 low (under 20%)'
                WHEN inactivity < 35 THEN '2 medium (20-35%)'
                ELSE '3 high (35% or more)' END AS inactivity_level
    FROM v_country_year
    WHERE year = 2022
)
SELECT inactivity_level, COUNT(*) AS countries,
       ROUND(AVG(life_expectancy), 1) AS avg_life_expectancy,
       ROUND((PERCENTILE_CONT(0.5) WITHIN GROUP (ORDER BY
life_expectancy))::numeric, 1) AS median_life_expectancy
FROM y
GROUP BY inactivity_level
ORDER BY inactivity_level;
```

inactivity_level	countries	avg_life_expectancy	median_life_expectancy
1 low (under 20%)	67	67.6	66.9
2 medium (20-35%)	68	73.9	74.5
3 high (35% or more)	57	75.7	75.4

Interpretation. Countries with low inactivity have the lowest life expectancy (67.6 years on average, median 66.9). Medium inactivity averages 73.9 years and high inactivity 75.7 years. This is the opposite of what one might expect, and it does not mean that inactivity prolongs life. Income level is a likely hidden factor, because Q11 shows high inactivity in rich countries. The inactivity estimates also come largely from self-reported surveys.

Q14. Years with the largest rise in average obesity

Question. In which five years did the average obesity of all countries rise the most compared with the previous year?

```
WITH yearly AS (
    SELECT year, AVG(obesity) AS avg_obesity
    FROM v_country_year
    GROUP BY year
), changes AS (
    SELECT year, avg_obesity,
           avg_obesity - LAG(avg_obesity) OVER (ORDER BY year) AS change
    FROM yearly
)
SELECT year, ROUND(avg_obesity, 2) AS avg_obesity, ROUND(change, 2) AS
change_points
FROM changes
WHERE change IS NOT NULL
ORDER BY change DESC
LIMIT 5;
```

year	avg_obesity	change_points
2020	22.26	0.45
2016	20.50	0.45
2022	23.15	0.45
2021	22.70	0.44
2017	20.94	0.44

Interpretation. In the five years with the largest increase, the average obesity of the 192 countries rose by 0.44 to 0.45 percentage points per year. The average was 20.50 % in 2016 and 23.15 % in 2022. The five values are almost identical, so the rise is steady and no

single year stands out. This smoothness probably comes from the way WHO models the series, so the data show a long-term trend and cannot show sudden changes in one year.

Q15. Import quality per source

Question. How many rows were read, loaded and not loaded for each source, and how many days have passed since the download?

```
SELECT ds.name AS source, b.started_at::date AS import_date,
       b.rows_read, b.rows_loaded, b.rows_rejected,
       ROUND(100.0 * b.rows_loaded / b.rows_read, 1) AS percent_loaded,
       CURRENT_DATE - ds.retrieved_on AS days_since_download
FROM import_batch b
JOIN data_source ds ON ds.source_id = b.source_id
ORDER BY b.batch_id;
```

source	import_date	rows_read	rows_loaded	rows_rejected	percent_loaded	days_since_download
WHO GHO: Insufficient physical activity, adults 18+, age-standardized	2026-11-12	41,262	13,248	28,014	32.1	1
WHO GHO: Obesity among adults, BMI >= 30, crude estimate	2026-11-12	28,350	13,248	15,102	46.7	1
World Bank: Life expectancy at birth, total (years)	2026-11-12	14,322	4,416	9,906	30.8	1

Interpretation. Only 31 to 47 % of the source rows were loaded, which is intended: the rest are aggregates, other age groups, other years and countries missing from one source (section 8). The column `rows_rejected` therefore counts all rows that were not loaded, not only invalid rows. The import date is the date of the last import run, and the files were retrieved one day earlier (days_since_download is 1 on the day the query was run).

Q16. Top life-expectancy countries and their risk profile (complex query)

Question. In 2022, which are the three countries with the highest life expectancy in each WHO region, and are their obesity and inactivity above or below the average of their region?

```
WITH y AS (
    SELECT * FROM v_country_year WHERE year = 2022
), r AS (
    SELECT who_region, AVG(obesity) AS avg_obesity, AVG(inactivity) AS
avg_inactivity
    FROM y GROUP BY who_region
), ranked AS (
    SELECT y.who_region, y.country, y.income_group, y.life_expectancy,
y.obesity, y.inactivity,
           RANK() OVER (PARTITION BY y.who_region ORDER BY y.life_expectancy
DESC) AS rank_in_region,
           CASE WHEN y.obesity > r.avg_obesity AND y.inactivity >
r.avg_inactivity THEN 'both above region average'
                WHEN y.obesity > r.avg_obesity THEN 'obesity above region
average'
                WHEN y.inactivity > r.avg_inactivity THEN 'inactivity above
region average'
                ELSE 'both at or below region average' END AS risk_profile
    FROM y JOIN r ON r.who_region = y.who_region
)
SELECT * FROM ranked WHERE rank_in_region <= 3 ORDER BY who_region,
rank_in_region;
```

who_region	country	income_group	life_expectancy	obesity	inactivity	rank	risk_profile
Africa	Algeria	Upper middle income	76.129	23.360	29.037	1	both above region average
Africa	Cabo Verde	Upper middle income	75.907	15.277	31.235	2	both above region average
Africa	Seychelles	High income	74.510	33.031	20.439	3	both above region average
Americas	Canada	High income	81.095	26.950	37.180	1	both at or below region average
Americas	Puerto Rico (US)	High income	79.427	40.825	41.552	2	both above region average

who_region	country	income_group	life_expectancy	obesity	inactivity	rank	risk_profile
Americas	Panama	High income	79.324	35.938	57.639	3	both above region average
Eastern Mediterranean	Qatar	High income	81.857	41.273	53.483	1	both above region average
Eastern Mediterranean	Bahrain	High income	80.992	34.569	38.614	2	obesity above region average
Eastern Mediterranean	Kuwait	High income	80.588	40.992	63.315	3	both above region average
Europe	Andorra	High income	84.016	19.399	27.209	1	inactivity above region average
Europe	Switzerland	High income	83.605	12.503	19.034	2	both at or below region average
Europe	Spain	High income	83.134	19.643	21.848	3	both at or below region average
South-East Asia	Maldives	Upper middle income	80.760	16.472	24.620	1	obesity above region average
South-East Asia	Sri Lanka	Upper middle income	77.300	10.213	37.182	2	both above region average
South-East Asia	Thailand	Upper middle income	75.291	14.347	28.872	3	both above region average
Western Pacific	Japan	High income	83.996	4.884	44.747	1	inactivity above region average

who_region	country	income_group	life_expectancy	obesity	inactivity	rank	risk_profile
Western Pacific	Australia	High income	83.200	31.455	25.466	2	obesity above region average
Western Pacific	Singapore	High income	82.895	15.024	23.175	3	both at or below region average

Interpretation. Long life expectancy does not require low obesity and low inactivity. Of the 18 countries, 9 are above their region's average in both obesity and inactivity, 3 only in obesity, 2 only in inactivity, and just 4 (Canada, Switzerland, Spain and Singapore) are at or below the average in both. Japan has the lowest obesity of the group (4.9 %) but inactivity above the Western Pacific average (44.7 %). In Africa all three leaders are above their region's average in both indicators, partly because African averages are low. The query combines a CTE for the regional averages, a window function for the ranking and a CASE expression for the classification. The averages count countries equally and the two WHO indicators are not measured in the same way, so the classification shows direction only.

Q17. Romania from 2000 to 2022

Question. How did Romania change in life expectancy, obesity and physical inactivity between 2000 and 2022?

```
SELECT year, life_expectancy, obesity, inactivity
FROM v_country_year
WHERE iso3 = 'ROU'
ORDER BY year;
```

year	life_expectancy	obesity	inactivity
2000	71.163	12.783	18.835
2001	71.161	13.472	19.363
2002	71.010	14.229	19.924
2003	71.310	15.047	20.517
2004	71.561	15.910	21.143

year	life_expectancy	obesity	inactivity
2005	71.912	16.812	21.802
2006	72.163	17.752	22.494
2007	72.566	18.750	23.218
2008	72.566	19.817	23.971
2009	73.310	20.942	24.751
2010	73.459	22.105	25.555
2011	74.410	23.277	26.383
2012	74.412	24.453	27.234
2013	75.063	25.637	28.106
2014	74.910	26.822	29.001
2015	74.912	27.999	29.917
2016	75.210	29.144	30.854
2017	75.210	30.264	31.806
2018	75.359	31.381	32.770
2019	75.607	32.492	33.747
2020	74.254	33.604	34.738
2021	72.810	34.662	35.742
2022	75.154	35.644	36.757

Interpretation. Between 2000 and 2022 Romanian life expectancy rose by 4.0 years (71.2 to 75.2), obesity by 22.9 percentage points (12.8 % to 35.6 %) and inactivity by 17.9 points (18.8 % to 36.8 %). Obesity and inactivity rise every year, by about 0.7 to 1.2 and 0.5 to 1.0 points, a smoothness typical of modelled estimates and not of yearly surveys. Life expectancy peaked at 75.6 in 2019, fell to 74.3 in 2020 and 72.8 in 2021, and recovered to 75.2 in 2022. The dip falls in the pandemic years, but the database holds no data on its causes.

12. Conclusion

The project turned three open files from WHO and the World Bank into a ten-table PostgreSQL database with 30,912 observations for 192 countries and the years 2000–2022. The source analysis found eight problems that made direct storage unsuitable: aggregates mixed with countries, several age groups, missing country names, a wide year layout, unmatched countries, different time coverage and different kinds of estimate. The design solves them with lookup tables, a generic observation table, an import script that cleans and maps the data, and constraints and triggers that keep the data valid. Eleven integrity tests and seventeen queries show that the database works.

Main findings

- Life expectancy rose in 189 of 192 countries between 2000 and 2022, and obesity rose in 191 (Q10).
- The longest life expectancy is found in Europe and the Western Pacific (Andorra and Japan, about 84 years), and the largest gains in Africa, up to 19.9 years (Q1, Q5).
- Countries with higher inactivity have higher average life expectancy (Q13), and the sign of the obesity correlation changes between regions (Q7). Both results point to other factors such as income, not to a direct effect.
- In the ten countries with the largest gaps, women are less active than men by 19 to 26 percentage points (Q8).

Limitations

- The WHO values are modelled estimates, and inactivity is age-standardized while obesity is a crude estimate, so the two are not directly comparable.
- All results are country-level averages. They are not weighted by population and say nothing about individual people.
- The queries show associations, not causes.
- The subset excludes territories missing in one source (for example 21 World Bank territories that WHO obesity lacks), the years outside 2000–2022, and all age groups except 18+.

Possible extensions

A population table would allow weighted averages. An `age_group` table would allow the other age groups of the inactivity data. Further indicators can be added with new rows in `indicator`, without changing the schema.

Sources

- World Bank: Life expectancy at birth, total (years)
- WHO GHO: Obesity among adults, BMI ⩾ 30, crude estimate
- WHO GHO: insufficient physical activity indicator group
- WHO data terms and conditions and CC BY-NC-SA 3.0 IGO
- Project repository